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# Effects of changing farming practices in African agriculture

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Information on the effects of changing agricultural management on crop and livestock performance is critical for developing evidence-based policies, investments, and programs. *Evidence for Resilient Agriculture* (ERA) v1.0.1 presents a dataset that harmonizes and aggregates 112,859 observations from 2,011 agricultural studies taken place in Africa between 1934 and 2018. The dataset includes information on the effect of 364 combinations of management practices and technologies on 87 environmental, social, and economic indicators of outcomes. Observations are geolocated and temporally tagged and thus can be linked to other datasets such as historical weather, soil properties, and road networks. ERA offers a new resource for understanding the impacts of changing farming practices under diverse environmental contexts, providing data to support strategic interventions aimed to enhance productivity, resilience, and sustainability of African agriculture.

## Background & Summary

*Which agricultural management practices or technologies work where in Africa?* This question lies at the forefront of development initiatives aimed at eradicating poverty, ensuring food and nutritional security, mitigating and adapting to climate change, and restoring degraded lands<sup>1,2</sup>. That is because the management practices or technologies (hereafter ‘technologies’) farmers use to produce crops and livestock determine yield, profitability, environment impacts, and more<sup>2</sup>. Hence, governments, the private sector, and program developers need information on the potential benefits or losses due to changing management practices to develop policies, programs, and products that support African farmers.

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| Search           | Publication years | Articles retrieved (N) | Passed abstract/title (N) | Passed full text (N) |
|------------------|-------------------|------------------------|---------------------------|----------------------|
| Initial search   | <2014             | 144,567                | 12,803                    | 785 (7,311)          |
| Recursive search | <2014             | 20,423                 | 1,477                     | 394                  |
| 2019 Update      | 2014–2018         | 25,455                 | 2,337                     | 972                  |
| ERA v1.0.1       | 1965–2018         |                        |                           | <b>2,011</b>         |

**Table 1.** ERA search and screening results. The initial search was global and resulted in 7,311 articles that passed through the two-stage screening process, of which 785 studies were conducted in Africa and are included in ERA v1.

Research efforts have generated a significant corpus of literature on agricultural technologies, with more than 500 meta-analyses conducted in the past decade to assess the efficacy of these practices for Africa agriculture<sup>3,4</sup>. Despite the volume of research, there remains a gap in the translation of these data into insights for agricultural programming<sup>5</sup>. This gap can be attributed to several factors: a narrow focus on a few technologies<sup>4</sup>, a lack of quantitative estimates of benefits<sup>6,7</sup>, publication delays that cause the findings to be released after they are needed for decision-making processes<sup>8</sup>, and/or misalignment with stakeholder questions<sup>9</sup>. Consequently, stakeholders' information needs are not fully met potentially leading to decision-making that is poorly supported by the available evidence<sup>10,11</sup>. Thus, there is a need for datasets that provide quantitative estimates of multiple agricultural technologies that can address a range of stakeholder questions<sup>12</sup>.

*Evidence for Resilient Agriculture* (ERA) v1.0.1 helps address the gap. Started in 2012, ERA was envisaged to evaluate the evidence base of Climate-Smart Agriculture (CSA)—that is, agriculture that delivers productivity, resilience, and climate change mitigation outcomes simultaneously<sup>13</sup>. However, the technologies included within ERA such as agroforestry, intercropping, and crop rotations, among many others are common features of agroecology, regenerative agriculture, nature-based solutions, ecosystem-based adaptation, sustainable land management, and other approaches as are many of the outcome indicators such as yield, net economic returns, soil organic carbon, land equivalent ratio, labour required, and more<sup>11,14–19</sup>. This means that ERA is relevant for various perspectives on development and is flexible to allow users to define 'effectiveness' or 'work' consistently with their worldview.

ERA is a comparatively large agricultural meta-dataset in terms of number of technologies, outcomes, and studies. ERA v1.0.1 includes data from 2,011 agricultural experiments that took place in Africa between 1934 and 2018. Together, these data compare how changing more than 364 combinations of agronomic, livestock, or tree management technologies affect more than 87 indicators of productivity, resilience, and greenhouse gas emissions and/or carbon stocks. The experiments were identified via *Web of Science* and *Scopus* and were evaluated against predetermined inclusion criteria: (i) location, (ii) technology and outcome relevance, (iii) data on both a new and conventional technology, and (iv) inclusion of primary data. Extracted data from the 2,011 studies ( $N = 112,859$  observations) include 135 fields describing each study's context, experimental design, management treatments (i.e., which are combinations of technologies used), and outcomes. This paper releases ERA v1.0.1. Future releases are planned. Meanwhile, we are developing the infrastructure to turn ERA into a living evidence synthesis to facilitate updates, expansion, and the broader research community's participation.

## Methods

Development of ERA has aligned to common practice for evidence reviews<sup>9</sup>. The methods described here update and expand on a previously released protocol<sup>1</sup>.

**Literature search.** Our initial search, conducted in 2014, queried the *Web of Science* and *Scopus* Databases for English language articles. Database queries consisted of key terms describing three components: technology, outcome, and geographic location (Table S1). Given the myriad outcomes of interest, we constructed distinct search strings for each category: productivity, resilience, and mitigation outcomes, as well as a key term related to barriers to adoption. The geographical aspect of each search string included low and middle income countries (LMICs) and specific regions of interest (e.g., 'Africa', 'Sahel', 'Amazon', etc.). Our methodology employed Boolean operators 'OR' and 'AND' to ensure the comprehensiveness of the search. This string for each technology category was run in both search engines for each of the outcome categories, 'productivity', 'resilience', 'mitigation', and additionally for 'barriers'. This approach returned 144,567 unique articles published between 1965 and 2013 inclusively (Table 1).

**Initial search and screening.** We used a two-stage screening strategy to determine the relevance of articles to our primary research question. Studies were excluded that did not: (1) include data on at least one technology and one outcome of interest identified *a priori* and covered in our search terms; (2) include data on both the new technology and a control (conventional or farmer's standard) technology; (3) take place in a LMIC as identified by the World Bank; or (4) report primary data from field trials. Modelling studies, and meta-analyses were excluded. Greenhouse experiments were only included when they were representative of a real-world growing system such as tomato production in polytunnels or presented outcomes related to greenhouse gas emissions due to the paucity of field data on climate change mitigation in Africa.

**Stage 1.** Title and abstract screening. Our team manually screened the 144,567 articles' titles and abstracts. To ensure consistency and reliability in reviewer decisions, preliminary rounds of screenings were conducted on subsets of 100 articles. Inter-reviewer agreement was tested and met minimum Cohen's kappa statistic of 0.6. Post reviewer calibration, each reviewer was allocated a technology theme (e.g., livestock) based on expertise, and proceeded to conduct title and abstract screening according to the inclusion criteria listed above. Of the total articles identified in the search, 12,803 (8.8%) were likely to meet the inclusion criteria.

**Stage 2.** Full text screening. Articles that passed the title and abstract screening were assessed in their entirety against the same criteria as the earlier abstract and title screening. The full text screening considered all criteria listed above and focused on outcomes, comparators, and primary data, which is less commonly described in titles and abstracts. The full text screening resulted in a final 2014 library of 7,311 references, comprising 57% of articles from Stage 1 and 5.1% of initial search results.

**Recursive search.** Due to human resource constraints and in response to stakeholder requests, in 2014, we focused efforts on studies conducted within Africa and augmented the library through a recursive search. This ‘recursive’ search—now commonly called ‘citation chasing’—was based on the reference lists from the 785 publications that occurred in Africa and had passed full-text screening. Articles identified through these searches underwent the same two-stage screening described above. This process yielded an additional 394 articles.

**2019 update.** The search and screening were repeated in 2019 to update the library for the years 2014 to 2018 for African countries adding an additional 972 articles to the corpus. Combining those articles with those identified during the previous search and screening efforts generated today’s ERA library of 2,011 studies published between 1965 and 2018 (Table 1).

**Data management.** Data were extracted from tables, text, and figures. Figures were digitized using available software, such as Graph Click (<http://www.arizona-software.ch/graphclick>) or Web Plot Digitizer (<https://apps.automeris.io/wpd/>). Each row of the dataset has a unique ID and represents a unique combination of article, site, comparison between new and common technology, commodity, outcome measure, and time period. For example, the same measurement of the same treatments at the same site in consecutive years (e.g., yield of treatment X at site Y in 2004 and 2005) would generate two observations. Studies of multiple treatments (e.g., three quantities of fertilizer) or outcomes measures (e.g., gross returns and soil carbon) contribute multiple observations to the dataset. Factorial studies conducted over multiple years measuring many outcome variables in many locations add large numbers of observations to the dataset. Below we describe the broad categories of variables compiled in the dataset. Descriptions of each variable can be found in Table S2.

**Record and bibliographic information.** Each row included the article’s bibliographic information, geographic location, environmental context, experimental design, treatment comparisons, and outcome indicator effects. In addition, ERA compiled the surname of the first author, the year of publication, and the journal abbreviation.

**Geographic location.** We collected country, site name paraphrased from study, and spatial coordinates when given. Location’s coordinates were verified in Google Maps, as they were often inaccurately reported. Enumerators also recorded a measure of spatial uncertainty. When authors reported decimal degrees and there was no correction required to the co-ordinates, then uncertainty was measured in terms of the value’s precision. When the location was estimated using Google Maps, the spatial uncertainty value was measured in terms of the precision of the site location description (e.g., a single farm or region) and the enumerator’s visual interpretation of land use at and near the coordinates. Observation’s geographic coordinates were collected to facilitate linking the data compiled in ERA to external databases, for example related to climatic and environmental factors not necessarily reported in the original study.

**Environmental context.** Information collected describes (i) growing season, (ii) climate, (iii) physical environment (elevation and slope), and (iv) soils. Growing season encoded rainfall as unimodal or bimodal and season start and end dates. Climate was typically reported as mean annual temperature and mean or total annual or seasonal precipitation. All reported were captured. Monthly amounts were summed to total and/or average seasonal and/or annual, as the reported data allowed.

Detailed soil information describing the site was also recorded. Soil classification and texture were recorded as reported. When articles report percentages of sand, silt and clay, soil texture was estimated based on USDA Soil Texture triangle and calculator [https://www.nrcs.usda.gov/wps/portal/nrcs/detail/national/home/?cid=nrcs142p2\\_054167](https://www.nrcs.usda.gov/wps/portal/nrcs/detail/national/home/?cid=nrcs142p2_054167)). Soil chemistry was recorded as a percentage, either as reported or converted from g/kg. Multiple depths, sites, and seasons were combined to match the detail for the respective outcome of the observation.

**Experimental design.** We recorded (i) the number of replicates of a treatment, (ii) the plot size harvested for yield measurements, converted to m<sup>2</sup> as the data allows, and (iii) the site type, e.g., research station, farmer field, or survey. When the number of replicates differed between comparisons (see below), the lowest number of replicates was recorded as a conservative measure.

**Treatment comparisons.** Each observation encodes a comparison between two treatments. Experimental treatments were coded according to the technologies labeled in the concept scheme (Table S3, Table S4). The following principles assisted in consistent comparisons of treatment outcomes across studies:

1. **New versus common treatment.** A new technology versus a control, which is typically farmers’ conventional technologies.
2. **Additive complexity.** An improved technology / set of technologies to a simpler option, e.g., agroforestry + fertilizer vs agroforestry alone.
3. **Agriculture to agriculture.** Agricultural systems are never compared to natural systems. For example, we can compare soil organic carbon in mulch versus no mulch systems, but not versus natural or semi-natural vegetation.

4. **Same implementation levels.** Comparisons were made within the same ‘level’ of implementation. For example, 40 kg N/ha was compared with other treatments using 40 kg N/ha but not 20 kg N/ha.
5. **All treatment details.** All major characteristics of the treatments were coded, including seed variety, tillage type, weed control, tree species, and chemical applications, among others, up to 13 labels per treatment.
6. **In-year comparisons.** Only comparisons of treatments and outcomes that occur in the same year or season were included. Residual effects, e.g., of phosphorus applied in year one with yield in year three, were not recorded.
7. **Include all possible treatments.** All treatments, and their component technologies, in a study were coded if they formed part of valid comparison.

**Outcomes.** Outcome measures, units, and products were extracted as reported. Outcome codes differentiated plant parts (grain vs stover) and products from the same species (milk versus meat). Oftentimes authors discussed many treatments but only reported a few. The ways in which the outcomes were reported in the papers, by year or aggregation across practices, dictated which data were extracted and how treatments were coded. Experimental Units (EU) described the species or product that were measured (e.g., maize grain).

## Data Records

The ERA v1.0.1 dataset includes 112,859 observations compiled from 2,011 articles found in *Web of Science* and *Scopus* (Fig. 1). The dataset, publicly available on GitHub, Zenodo<sup>20</sup> and Dataverse<sup>21</sup> consists of an R object in the ERAg package and Microsoft Excel workbook, respectively. It includes 135 fields, described in methods (Table S2). The package and file include the codebooks for technologies, outcomes, and products.

The dataset provides an extensive resource on the effects of changing crop, livestock, and tree management practices on agricultural, social, and environmental outcomes. Data are available from 42 countries, but research efforts have been unevenly distributed, with notable hotspots in certain countries (Fig. 1a). The countries most represented are Nigeria ( $N=397$  studies, 19.7%), Kenya (218, 10.8%), Ethiopia (212, 10.5%), South Africa (151, 7.5%), and Zimbabwe (134, 6.6%). Meanwhile the database contains no or few studies from Angola, Central African Republic, the Democratic Republic of Congo, Liberia, Gabon, Chad, and Namibia.

The ERA dataset contains information on a diverse range of agricultural products including both crops and livestock (Fig. 1b). Most studies focus on cereals (1262 studies), maize (916), rice (97), wheat (99), sorghum (141), and millet (116). However, there is also a significant amount of information on legumes (504) and other starchy staples such as cassava (108), potatoes (30), and vegetables (117). ERA also include data on livestock. About 25% of the studies describe livestock management (442), including more than 200 studies on goats and sheep and 80 studies on cattle.

ERA categorizes practices at three levels, with increasing specificity (Table S3). At the highest and broadest level of aggregation (‘themes’), ERA includes information on 36 technology categories such as agroforestry, inorganic fertilizers, crop rotation, and improved feeding of animals. Nutrient and soil management practices are most well represented in the dataset, and are the focus of 1284 and 1044 studies, respectively. Other important themes include agroforestry (426 studies), water management (449 studies), and livestock management including practices like changing feeding practices (442 studies). Nearly 500 studies examine crop diversification including intercropping or rotations. However, studies were coded at greater specificity (‘practices’ and ‘sub-practices’) such as alley cropping with nitrogen fixing tree, or improved feeding with leguminous fodder. ERA captures information on 113 and 1,285 of sub-practices used alone and in combination, respectively.

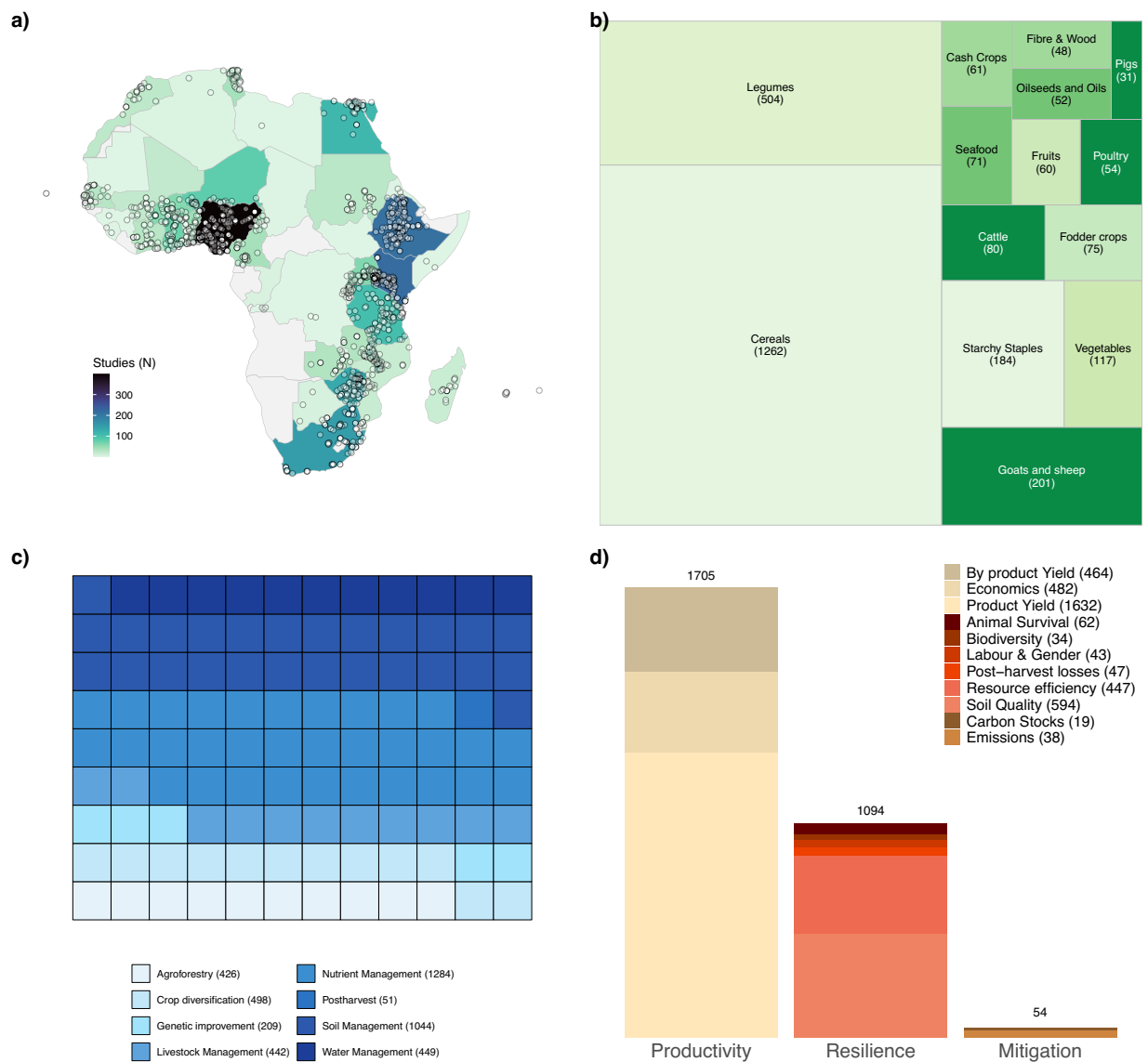
ERA compiles data on the impacts of technology use on 87 indicators of productivity, resilience, and climate change mitigation outcomes (Fig. 1d). Productivity is studied most extensively (1705 studies) and includes both yield and economic performance indicators such as net returns. Social and environmental proxies for resilience such as resource use efficiency and soil organic carbon are included in 1094 studies. Climate change mitigation, although studied less often, with only 54 studies, includes both greenhouse gas fluxes and soil carbon stock changes. It is important to note that in many cases single studies contain multiple outcomes allowing ERA to be used to look at synergies and trade-offs among outcomes.

The geographic and topic distribution of studies provide valuable insights into historical research efforts and highlight future research needs. This distribution can identify over- and under-researched agroecologies, commodities, practices, or outcomes, particularly when coupled with information on the importance of commodities to food security, poverty alleviation, or other development and environmental impacts. It is important to note however that a lack of data in any specific location, practice, or outcome does not necessarily indicate a complete absence of information in the literature. Instead, it suggests little or limited available data, and especially data that meets the criteria that required a control in the studies analysed.

## Technical Validation

Meta-analyses inherently require decisions regarding search parameters, such as the selection of keywords, the establishment of inclusion criteria, and the choice of analytical methods. These decisions can significantly influence the data compiled and the study’s conclusions, thus necessitating an evaluation of the data compilation to the extent possible to contextualize the results.

The process of search and screening represents a potential source of error, primarily due to variability in human judgement. To mitigate this, we implemented rigorous training and calibration for our enumerators. This process involved pairs of enumerators independently assessing the same articles and reconciling differences under the guidance of a project lead. Calibration was deemed satisfactory when inter-rater reliability, measured by the Kappa statistic, consistently exceeded 0.6—reflective of the standard in meta-analytical research.



**Fig. 1** Systematic map of ERA. **(a)** Geographic locations of studies, **(b)** Distribution of products included by number of studies, **(c)** Distribution of technology groups for crop and livestock by number of studies with each square equal to ~40 papers, **(d)** Number of studies including outcome indicators related to farm productivity, resilience, and climate change mitigation.

Data extraction accuracy is another concern. To mitigate this, enumerators received extensive training over a 4 to 6-weeks period. The training followed a detailed manual that explains the concepts and provides examples of the types of data, experiments, and complexities likely to be encountered. Following training, ongoing support and daily meeting helped reconcile any extraction concerns. Slack was then used to enable rapid feedback for any questions among peers and with project leads. Despite the precautions, errors still occur. Thus, we adopted the Four Eyes Principle. Each article's extraction was checked by a second enumerator or a project lead, and in some cases both. Errors were rectified to achieve a high standard of data integrity.

This commitment to accuracy in data extraction lays a strong foundation for the subsequent validation of ERA. The validity of ERA can be substantiated through comparative analysis with the results from other meta-analyses. Crop yield, which represent 38.8% of ERA's data (Fig. 1), is also among the most investigated outcomes in meta-analysis. We analysed ERA's crop yield data using log response ratios and effect sizes<sup>22,23</sup>. Critically, the calculated effect sizes were in the range of these other studies (Table 2). These results indicate that ERA generates results expected by other, more specific, assessments of the literature and suggest ERA's approach delivers quality results. Furthermore, since the effect size calculations integrate all potential sources of error together, the validation results provide a robustness check on ERA processes broadly, enhancing confidence in the dataset's overall reliability.

Exact replication of results across meta-analyses is not expected due to unique protocols, search strategies, and analytical decisions employed by individual research teams. Consequently, the datasets and results can exhibit considerable variability as demonstrated by meta-analyses comparing the effects of organic farming on yield<sup>4</sup> and agroforestry on carbon sequestration<sup>24</sup>. This variability is also reflected in our validation of ERA. For specific



| Practice                          | Existing literature |            | ERA         |                     |
|-----------------------------------|---------------------|------------|-------------|---------------------|
|                                   | Studies (N)         | Effect (%) | Studies (N) | Effect (%)          |
| Agroforestry                      | 61                  | 81         | 176         | 22.9 (12.1–34.6)    |
| Intercropping                     | 58                  | 23         | 35          | 35.9 (31.6–40.2)    |
| Conservation agriculture          | 79                  | 8.4        | 18          | 20.8 (7.4–35.9)     |
| Reduced tillage                   | 20                  | 0          | 141         | –0.9 (–5.5–3.8)     |
| Organic fertilizers               | 57                  | 60         | 192         | 63.4 (55.2–72.1)    |
| Inorganic fertilizers             | 57                  | 84         | 524         | 59.4 (54.5–64.4)    |
| Inorganic and organic fertilizers | 57                  | 114        | 114         | 117.4 (101.7–134.3) |
| Green manure                      | 60                  | 63.7       | 46          | 15.6 (5.1–27.2)     |
| Crop residue retention            | 112                 | 2.6        | 213         | 26.3 (25.0–37.5)    |
| Improving feeding*                | 25                  |            | 167         | 38.8 (28.0–50.5)    |
| Feed substitution                 |                     |            | 242         | 0.7 (–1.5–2.9)      |
| Improved breeds*                  | 25                  |            | 24          | 63.1 (32.3–101.1)   |

**Table 2.** Comparing the effect changing technologies has on crop and livestock product yield between ERA and other meta-analyses of studies occurring in Africa. Effect reports mean for the existing literature and the mean and 95% confidence interval for ERA. Intercropping results based on land equivalent ratio (LER) and not yields to account for both crops. References for existing literature: agroforestry<sup>26</sup>; intercropping<sup>27</sup>; conservation agriculture and reduced tillage<sup>28</sup>; organic, inorganic, and combined fertilization<sup>29</sup>; and green manure and crop residue retention<sup>30</sup>. Improving feeding and improved breeds significantly affects milk production (\*<sup>31</sup>) but the analytical approach in that study differs from that used in ERA and so does not present a directly comparable result.

| User                                | Practices | Outcome                       | Analysis                                                                           | Evidence                             |
|-------------------------------------|-----------|-------------------------------|------------------------------------------------------------------------------------|--------------------------------------|
| <b>Policy support</b>               |           |                               |                                                                                    |                                      |
| Kenyan County Governments (3)       | 11        | Productivity, soil management | Effects of new practices under various environmental conditions                    | Lamanna <i>et al.</i> <sup>32</sup>  |
| <b>Development programming</b>      |           |                               |                                                                                    |                                      |
| GIZ                                 | 6         | Yield                         | Yields under climate stress                                                        | Jalango <i>et al.</i> <sup>33</sup>  |
| Global Center on Adaptation         | 19        | Yield time series             | Estimate production risks                                                          | GCA <sup>34</sup>                    |
| <b>Investment design</b>            |           |                               |                                                                                    |                                      |
| National governments (4) & partners | 50+       | Yield, costs and returns      | Input data for project cost-benefit analyses                                       | World Bank <sup>35,36</sup>          |
| IFAD                                | 20        | GHG emissions, soil carbon    | Estimate the climate impacts of development programs                               | Richards <i>et al.</i> <sup>37</sup> |
| CGIAR                               | Unknown   | Unknown                       | Input data for commercial banks                                                    | Burra <i>et al.</i> <sup>38</sup>    |
| <b>Research prioritization</b>      |           |                               |                                                                                    |                                      |
| The Nature Conservancy & partners   | 12        | Biomass and soil carbon       | Carbon stock changes with agroforestry                                             | Cook-Patton <i>et al.</i>            |
| 4 per 1000                          | 35        | Yield and soil carbon         | Synergies between soil carbon accumulation and yield                               | Soussana <i>et al.</i> <sup>39</sup> |
| CGIAR                               | 16        | Yield                         | Project benefits of research investments                                           | Jarvis <i>et al.</i> <sup>40</sup>   |
| <b>Financial services</b>           |           |                               |                                                                                    |                                      |
| ACRE Africa                         | NA        | Yield                         | Predict yield changes with and without practices for credit and insurance products | Discussed                            |
| Greenfi                             | NA        | Yield                         |                                                                                    |                                      |

**Table 3.** Diverse applications of ERA in policy, programming, research, and investment design.

technologies, ERA includes more studies than most related meta-analyses but not all. Differences can be attributed to definitional discrepancies, inclusion decisions (e.g., grey literature), and analytic criteria for what constitutes a valid comparison. These observations further underscore the significance that these decisions have on research results.

## Usage Notes

**Concept scheme.** ERA uses a hierarchical concept scheme to classify practices, outcomes, and products. Related concepts are nested within similar concepts. ERA's concept scheme is unique and was born from experience with the literature and deep engagement with stakeholders about their needs. This structure allows the data to be aggregated and disaggregated to respond to specific user questions at different levels of specificity and to combine across studies when using dimensionless analysis approaches such as response ratios and effect sizes. Multiple agricultural ontologies have been developed since ERA was conceived in 2014. In 2020, ERA concepts were mapped and aligned to other agricultural semantic ontologies and thesauri, including FAO's AGROVOC

and CGIAR's AgrO, to increase interoperability and future dataset expansion. The aligned concepts are included in the dataset.

**Interoperability.** ERA is designed to connect to other publicly available datasets. This allows it to reuse the data contained within to answer new user-specific questions. There are at least four ways to link ERA to other datasets, by: geographic coordinates, growing season dates, species data, and practice data.

**Use cases.** ERA was co-designed with various stakeholders, including scientists, policy makers, development partners, and investors. They contributed to identifying the relevant questions, analyses, and data visualization possibilities that provide actionable insights for different users. So far, six primary use cases have been developed or discussed with partners (Table 3). Much of the early engagements focused on generating context-sensitive analyses of costs, benefits, and risks for investment design and policy formulation. Yet, additional use cases of ERA's analytical potential have emerged from discussions with stakeholders. For example, farm management data under different environmental, social, and economic conditions lends itself to other decision and planning processes, including advisory service, design of targeted financial products, monitoring and evaluation of development projects, and prioritization of research agendas. These diverse applications underscore ERA's versatility and its utility in enhancing decision-making across various contexts.

**Data analysis.** ERA is made available with accompanying R packages ('ERAg' and 'ERAgON'). These packages contain functions for conducting the basic meta-analysis using response ratios and effect sizes as well as extracting treatments with multi-year observations or all the unique treatments in the dataset. The principles described above determine which data were extracted and included in the dataset. For example, ERA v1.0.1 did not encode comparisons that captured substituting one fertilizer source for another. Nor did we extract data to compare among new technologies except in terms of multiple-technology bundles, though subsequent releases will include such information. However, because each treatment was coded separately, extracting individual treatments without considering the comparison between technologies is possible. This unique treatment subset would not capture treatments in the studies where the comparisons were not the target of ERA's data extraction. These decisions determine ERA's relevance for some uses. Nevertheless, we expect that the value of the subset extraction and the availability of consistent cross-technology data outweighs imprecision concerns for most research questions and use cases but should be done with caution. The authors welcome inquiries about these or other limitations.

### Code availability

The code used to generate the dataset and run basic analyses is compiled in the ERAg R package<sup>25</sup> using R 4.2.1 (R Development Team) and may be downloaded from GitHub <https://github.com/EiA2030/ERAg>. This beta version package is actively being developed, so new functions and improvements should be expected.

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## Author contributions

T.S.R. conceived the project; T.S.R., P.S., N.J., C.L. developed the methodology; T.S.R., P.S., C.L. developed formal analysis; P.S. developed the R packages with support from T.S.R. and N.J.; All authors curated data; T.S.R. wrote the original draft; All authors reviewed and edited the draft; T.S.R. acquired funding.

## Competing interests

The authors declare no competing interests.

## Additional information

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